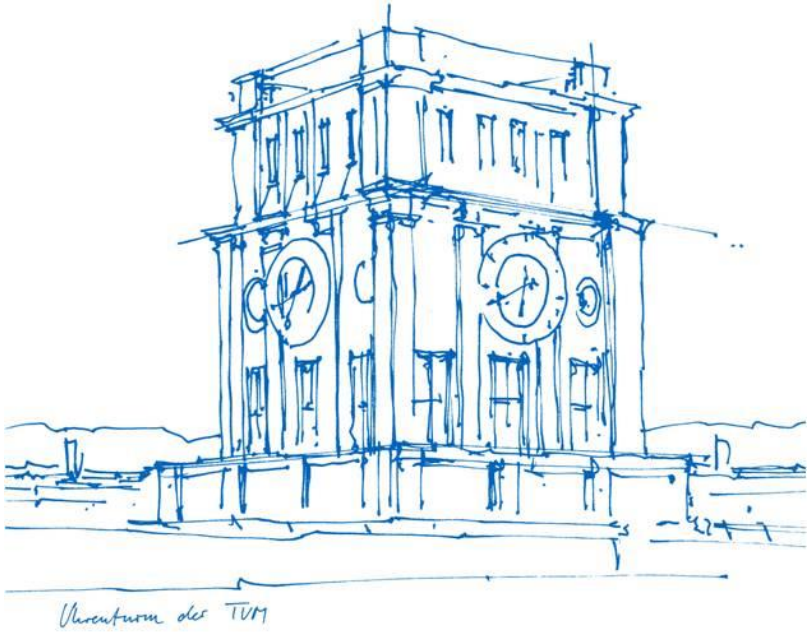
Energy Informatics Lab

Assignment 1 Wind Power Prediction

Group B

Technical University of Munich

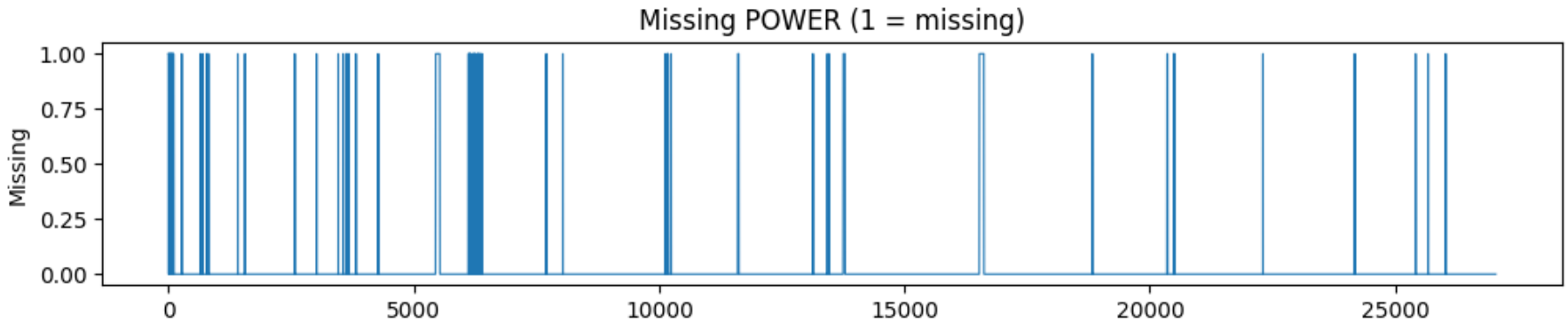


Data Cleaning

- Inspect train data with .info()
  - First row irrelevant -> remove
  - TIMESTAMP is object type -> pandas to_datetime
    - Easier sorting of hours, days, months

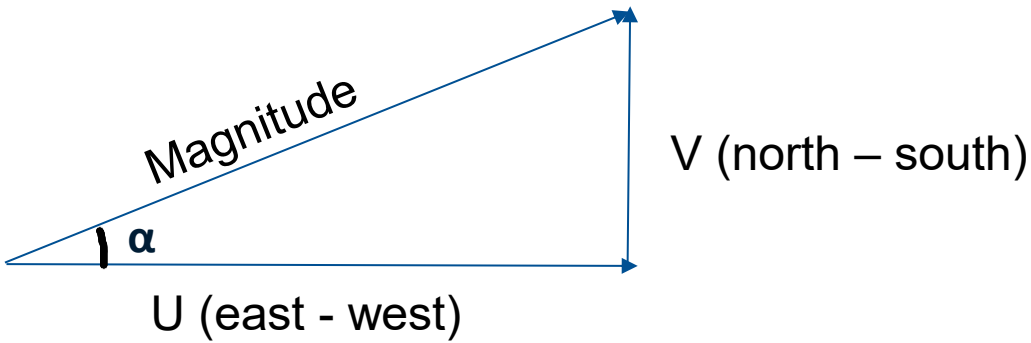
Data Cleaning – missing values

- POWER attribute has missing values
  - Persistance fill
  - 5 hour moving- average fill
  - Hourly climatology



Data Cleaning – coordinates

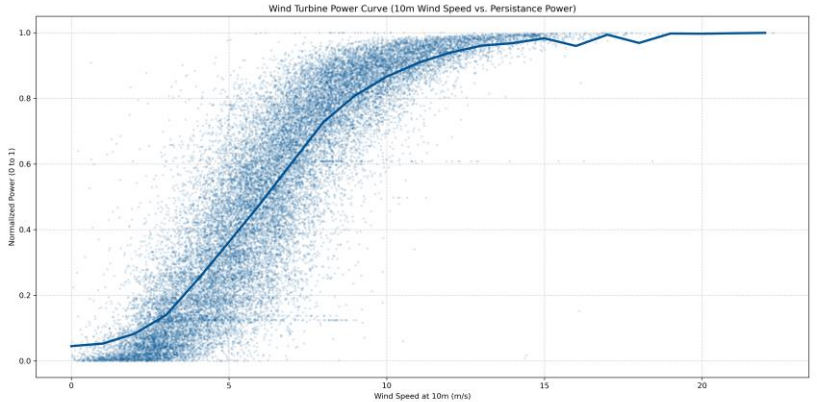
- Change from cartesian u-v coordinates to polar
  - New features
    - Wind speed (magnitude)
    - Wind direction (angle)

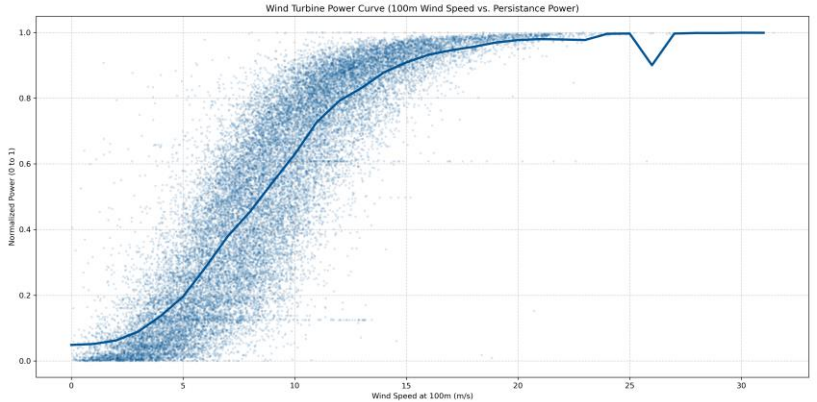


Data Exploration

- „S“-shaped power curve
- Similar shape at both heights

➢ Drop 10m data
➢ Categorical feature (cut-in, ramp-up, max-power)

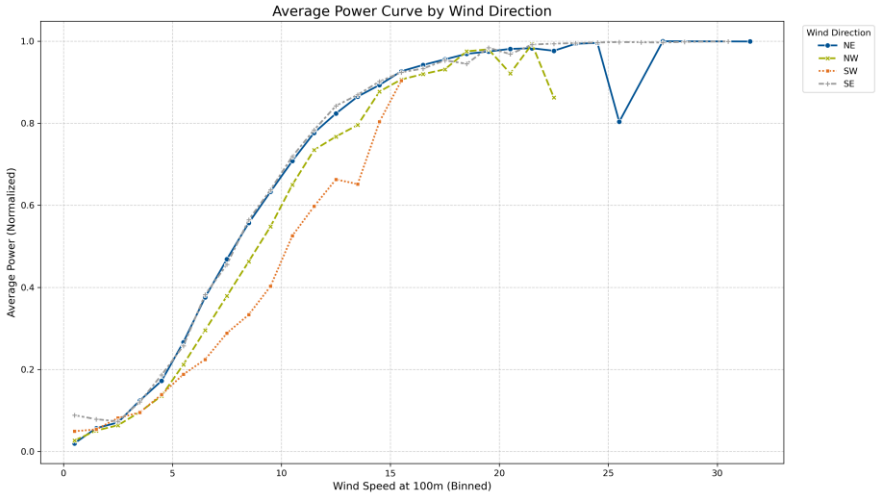




Data Exploration

- Power curves for different wind directions
- Power for wind to NW and SW lower
- Possible explanation: Geographical obstruction

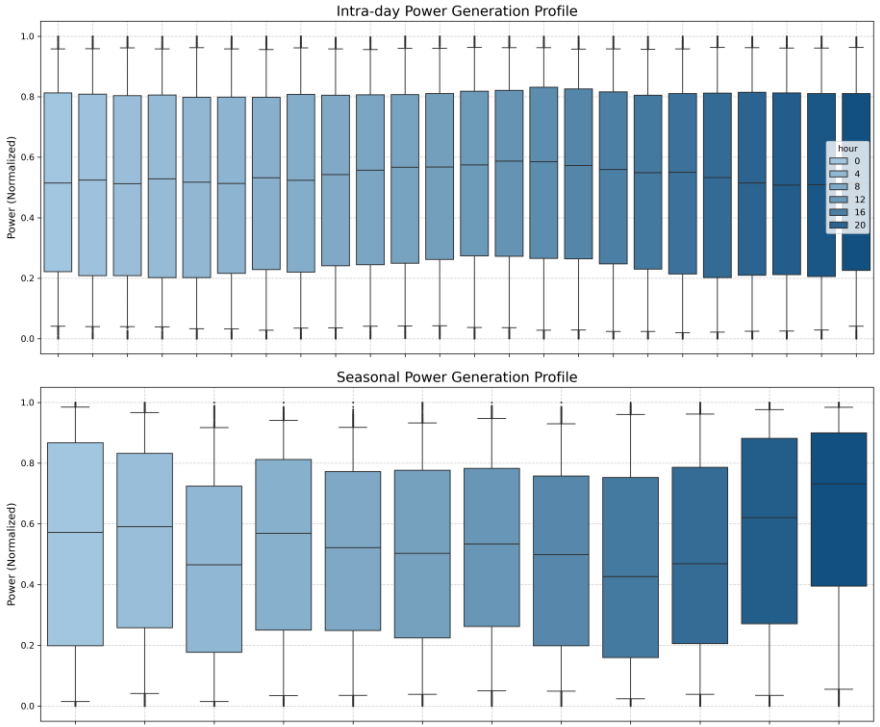
➢ Direction has impact on power
➢ Categorical feature (NE, NW, SW, SE)



Data Exploration

- Weak intra-day pattern
- Distinctive seasonal pattern

➢ Month has significant impact on average power
➢ Cyclical features for hours and months



Model Training

- XGBoost (Gradient boosted decision trees)
- Random Forest Regressor
- Linear Regression

- Mean R^2 scores (MAE shows a similar trend):

Model \ Data Fill	Persistence	Moving Average	Hourly Climatology
XGBoost	0.769	0.768	0.768
Random Forest	0.762	0.761	0.761
Linear Regression	0.752	0.752	0.751

Model Evaluation

- Further fine tuning of XGBoost hyperparameters using RandomizedSearchCV
- Achieved best R^2 score of **0.780** on the training data
- Preprocessed the WeatherForecastInput_A data and ran it through the fine-tuned model:
  - Final R^2 score: **0.857**
  - Final MAE: **0.095**
- Perhaps improved accuracy due to no missing data in the test set

Backup Slides

- Higher power for winds to E
- Possible explanations
  - Wind to W not strong enough to reach max power
  - Geographical obstruction

